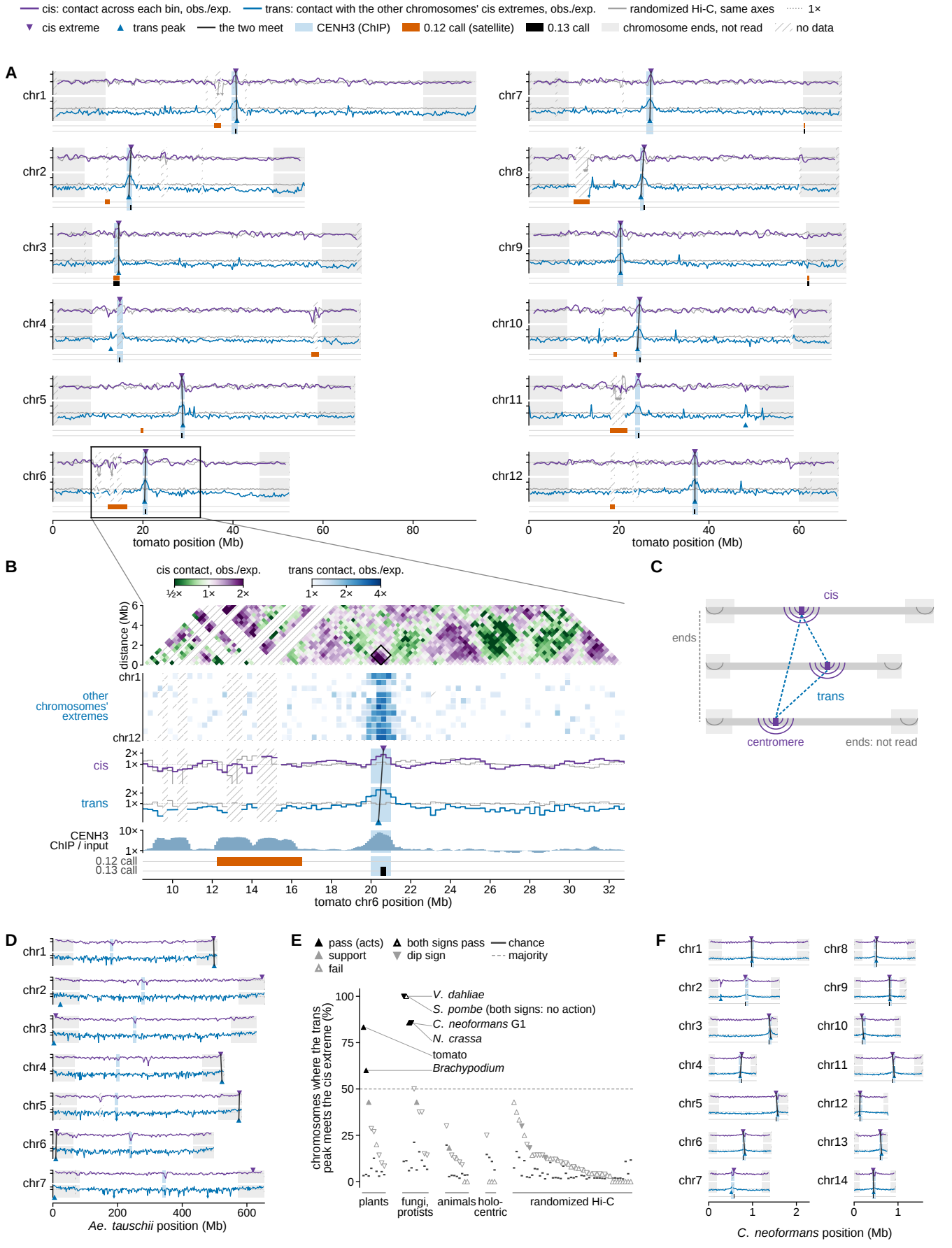

cis: contact across each bin, obs./exp.
trans: contact with the other chromosomes' cis extremes, obs./exp.
randomized Hi-C, same axes
1×
cis extreme
trans peak
the two meet
CENH3 (ChIP)
0.12 call (satellite)
0.13 call
chromosome ends, not read
no data
A
chr1
chr2
chr3
chr4
chr5
chr6
chr7
chr8
chr9
chr10
chr11
chr12
tomato position (Mb)
B
cis contact, obs./exp.
½× 1× 2×
trans contact, obs./exp.
1× 2× 4×
distance (Mb)
other chromosomes' extremes
cis
trans
CENH3 ChIP / input
0.12 call
0.13 call
tomato chr6 position (Mb)
C
cis
ends
trans
centromere
ends: not read
D
Ae. tauschii position (Mb)
E
pass (acts)
support
fail
both signs pass
dip sign
chance
majority
V. dahliae
S. pombe (both signs: no action)
C. neoformans G1
N. crassa
tomato
Brachypodium
chromosomes where the trans peak meets the cis extreme (%)
plants
fungi, protists
animals
holo-centric
randomized Hi-C
F
C. neoformans position (Mb)

Two readouts of one Hi-C library find centromeres that no repeat marks: where the contact across a stretch of chromosome peaks and the contact with the other chromosomes' peaks rises at the same place, that place is the centromere. No annotation is needed.

A In tomato, the contact across each chromosome's interior peaks once, and the contact with the other chromosomes' peaks rises at the same place on 10 of 12 chromosomes, on CENH3. A randomized library read the same way shows neither. The calls move from satellites onto CENH3.

B On tomato chr6, the two flanks of the centromere contact each other more than expected (diamond), and the same bins contact the centromeric peaks of all 11 other chromosomes (blue column): two contact patterns at one CENH3 domain, 3.5 Mb from the satellite the old call used.

C Centromeres fold locally and cluster in the nucleus; chromosome ends do both too, so only the interior of each chromosome is read.

D In *Ae. tauschii*, read with the ends kept, both readouts peak at chromosome ends and agree with each other away from CENH3; with the ends excluded no agreement remains.

E Across Hi-C libraries the agreement far exceeds chance in tomato, *C. neoformans*, *Brachypodium*, *N. crassa* and *V. dahliae*; in randomized libraries it never reaches a majority.

F In the fungus *C. neoformans*, the same test finds 12 of 14 centromeres at 10-kb resolution, where the earlier release called none.

Terms. Cis profile: per bin, the balanced Hi-C contact crossing it (4 bins on each side) over the chromosome's mean contact at those distances. Cis extreme: its highest bin outside the terminal eighths (shaded; ends fold and cluster whatever the centromere). Trans score: per bin, the mean observed/expected contact with the other chromosomes' cis extremes (± 4 bins), never its own. Meet: the trans maximum lies within 4 bins of the cis extreme and not on a contact hub, rDNA or telomere; chance: that zone's share of the chromosome. Pass: a majority meet and P ≤ 0.01 (Poisson-binomial, doubled for the two signs, peak or dip); support: beyond chance on fewer. Only pass moves calls.

A *S. lycopersicum* Heinz 1706, 250-kb bins, all 12 chromosomes at one scale (y ticks at 1× and 2×; hatched, unmappable). Gray: the same two readouts of a randomized library (contacts re-paired, coverage kept). Values beyond the axis are drawn at its edge with carets: the randomized cis profile beside unmappable stretches on chr8 and chr11 (up to 4.6×) and one tomato trans bin (chr8). The trans peak meets the cis extreme on 10/12 chromosomes (0.39 expected by chance; P 1.2×10^{-13}); all 12 cis extremes lie within 1 Mb of CENH3, and on chr4 and chr11 the trans maximum sits on a contact hub. The randomized library meets on 0/12 (peaks) and 1/12 (dips): fail. cenfuse 0.13 moves 9 low-confidence satellite calls to the cis extreme and confirms 1: right calls rise from 1/12 (0.12) to 10/12, none confidently wrong. The calls on chr7 and chr9 lie in a terminal eighth, which the line cannot judge, and stay wrong; their cis extremes lie on CENH3.

B Tomato chr6, the cis extreme ± 12 Mb. Top: observed/expected contact within the chromosome, each pair of bins drawn above its midpoint, up to 6 Mb apart (hatched, unmappable); the diamond is the crossing box at the cis extreme. Rows: the chromosome's contact with each other chromosome's cis extreme (± 4 bins), chr1 to chr12; their mean is the trans score. Gray lines: the randomized library. The trans peak lies 1 bin from the cis extreme, inside the CENH3 domain; CENH3 ChIP over input (100-kb bins, the truth rule's own track) peaks there at 7.9×. The 0.12 call sat on a satellite array 3.5 Mb away, which also carries CENH3 ChIP signal (4.7×, 59% of the peak) but no cis or trans extreme; 0.13 calls the cis extreme.

C Flat schematic, not data: purple arcs, contacts crossing a centromere (cis); blue dashes, centromeres of different chromosomes in contact (trans); gray, the same at chromosome ends.

D *Ae. tauschii* AL8/78, 2-Mb bins, all 7 chromosomes read with the ends kept (symbols as in A; the trans score is made from the other chromosomes' whole-chromosome extremes; CENH3 domains drawn at least 0.6 mm wide). The cis maxima lie in a terminal eighth on 7/7 chromosomes and the trans peak meets them on 4/7 (P 2.8×10^{-5}, one sign), all at chromosome ends (on chr3 the trans peak sits on the telomere itself, which never counts); 0/7 lie on CENH3. Read outside the terminal eighths, 0/7 meet. Tomato is unchanged by the rule (10/12 with the ends kept).

E Every Hi-C library where cenfuse 0.13 ran the test, one point per contact map (78; arms that reuse a map drawn once); up, the peak sign; down, the dip sign (the sign with the lower P). Bars: chance, the expected number of chromosomes meeting, re-read from all 78 matrices. Pass: tomato (10/12), *C. neoformans* G1 (12/14), *Brachypodium* (3/5), *N. crassa* (6/7, dips), *V. dahliae* (8/8, dips). Support: barley (3/7), *C. neoformans* in anaphase (6/14), rhesus (4/22). *S. pombe* passes with both signs (3/3): sign unknown, no action. Randomized: 0 of 46 pass, 3 reach support (soybean 6/20, rhesus 4/22, human 3/23), because both readouts share the library's coverage: hence action only at a majority. Holocentric genomes, without one centromere, all fail.

F *C. neoformans* H99 in G1, 10-kb bins, all 14 chromosomes (as in A without the randomized partner; black, 0.13 calls; light blue, CENP-A, drawn at least 0.6 mm wide). The trans peak meets the cis extreme on 12/14 (P 1.4×10^{-13}); right calls rise from 0/14 (0.12 made none) to 12/14, none confidently wrong; the two misses lie 19–20 kb beyond CENP-A. The Hi-C engine's own prior-free peaks lie at chromosome ends on 14/14, so only the interior read finds these centromeres.

Exceptions. Potato's whole-chromosome dips lie on CENH3 on 9/11 chromosomes, but its trans contacts do not confirm them (1/12, fail): the line misses there. Calls the line places or moves stay low confidence. Example: among the 8 tomato chromosomes whose trans peak meets the cis extreme and that 0.13 calls right, chr6 has the extreme height (1.82×; median 1.93×) and CENH3 width (1000 kb; median 1050 kb) nearest the median (each scaled by its median absolute deviation; next, chr12). Its extreme lies inside the CENH3 domain, outside the terminal eighths, with no unmappable bin within 4 bins. It is not the strongest case.

Analysis: asc-oulab:/data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12 · Script: /data2/chris/centromere_benchmark/v2/v3/v4/v5/v6/v7/v8/v9/v10/v10.1/v10.2/v10.3/v10.4/v11/v12/figures/fig_cis_trans/fig_cis_trans.py · 2026-11-07